Cost-Aware Architecture Search and Uncertainty Estimation for Lightweight 3D U-Nets on BraTS 2020

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ABSTRACT

3D U-Nets for glioma segmentation are routinely trained at a size chosen by convention rather than by measurement, and their predictions are delivered without any statement of confidence. This work addresses both. First, five hand-designed 3D U-Nets spanning 124K to 19.2M parameters are trained under an identical pipeline on BraTS 2020, establishing that the conventional 19.2M-parameter configuration is substantially over-parameterised for the task. Second, a cost-aware architecture search maximising Dice penalised by the logarithm of parameter count and FLOPs is run over channel width, encoder depth and residual depth. Rather than fixing the two penalty weights by intuition, the search is executed over their full 5×5 grid and the resulting sensitivity surface is used to identify the regime in which the objective still discriminates between architectures. The selected model, UNet-AutoBest, uses 428,553 parameters and reaches 0.852 mean region Dice against 0.857 for the 19.2M-parameter baseline — a 45× parameter reduction for a 0.005 Dice cost. Replacing every convolution with a depthwise-separable factorisation reduces this to 100,477 parameters (191× smaller than baseline, 9× fewer FLOPs) at 0.813 Dice. Third, Monte Carlo dropout on the compressed model produces per-voxel predictive entropy maps evaluated under the QU-BraTS protocol, yielding a score of 2.650 out of 3 and an expected calibration error of 0.011. Voxel-level uncertainty correlates with voxel-level error ($\rho = 0.22$), but the case-level correlation is weak ($\rho = 0.11$), indicating that these maps support intra-scan triage rather than automatic rejection of whole studies. All code, experiment records and configuration are released under MIT.

1 Problem and scope

Glioma segmentation on multi-modal MRI is dominated by U-Net variants, and the field's default answer to a new dataset is a five-level encoder with 32 base channels. That configuration is rarely justified by measurement on the dataset at hand. It is also expensive: 19.2M parameters and 49.7 GFLOPs per 128^3 forward pass is a poor fit for the settings where automated segmentation would actually help — edge deployment, batch screening of large cohorts, or interactive clinical tools that must respond in seconds.

This project asks two questions. How small can a 3D U-Net be on BraTS 2020 before accuracy degrades meaningfully? And, since a smaller model is a less certain model, can that model report where it is likely to be wrong? The first is treated as a constrained search problem rather than a manual sweep; the second is evaluated with the BraTS challenge's own uncertainty protocol rather than a qualitative eyeball test.

2 Data and experimental protocol

All experiments use the BraTS 2020 training set (369 cases, four co-registered modalities: FLAIR, T1, T1ce, T2). A single case-level 80/20 split with seed 42 gives 295 training and 74 validation cases, written to disk so that every run — anchors, search candidates and the final models — is evaluated on exactly the same held-out cases. There is no patient overlap between the two sides.

Preprocessing is fixed across every experiment: RAS reorientation, 1 mm isotropic resampling, non-zero z-score intensity normalisation per channel, and foreground cropping. Training samples are 128^3 crops drawn with balanced positive/negative sampling, augmented with axis flips, 90° rotations and mild affine jitter. The loss is a combined Dice and cross-entropy objective; optimisation uses AdamW with cosine annealing, a maximum of 100 epochs, and early stopping at patience 20 on validation Dice. Validation uses sliding-window inference at the training ROI.

Holding this pipeline constant is the point: every difference reported below is attributable to architecture alone, not to a tuning advantage.

2.1 Two Dice conventions

Two averages appear in the literature and both are reported here to avoid ambiguity. Mean class Dice averages the three raw label classes (necrotic core, peritumoural oedema, enhancing tumour); this is what the training code optimises and logs. Mean region Dice averages the nested BraTS regions — whole tumour (WT, labels 1∪2∪3), tumour core (TC, labels 1∪3) and enhancing tumour (ET, label 3) — and is the quantity comparable to published BraTS results. Region Dice is always the higher of the two because WT is an easy union. Unless stated otherwise, Dice in this report means region Dice.

3 Results

Model	Params	GFLOPs	WT	TC	ET	Mean region	Mean class	Time
UNet-Baseline (32,64,128,256,512)	19,225,897	49.66	0.903	0.873	0.793	0.857	0.762	6.2 h
UNet-Lite (16,32,64,128,256)	4,811,129	14.07	0.907	0.875	0.809	0.864	0.766	7.2 h
UNet-UltraLite (16,32,64,128)	603,649	9.42	0.902	0.863	0.799	0.854	0.756	6.8 h
UNet-Efficient (16,32,64,128), no res	489,322	4.57	0.896	0.854	0.774	0.841	0.739	7.6 h
UNet-Tiny (8,16,32,64)	123,706	1.49	0.877	0.819	0.738	0.811	0.686	7.4 h
UNet-AutoBest (24,32,80) ← searched	428,553	24.60	0.901	0.869	0.786	0.852	0.757	4.0 h
UNet-AutoBest-DSConv ← compressed	100,477	5.54	0.877	0.819	0.744	0.813	0.707	4.6 h

Table 1. Validation performance on the held-out 74 cases. The top five rows are hand-designed anchors; the shaded rows are the searched architecture and its depthwise-separable compression. Training time is wall-clock on a single lab GPU. Full per-run records are in `results/`.

The anchor sweep alone answers the first question. UNet-Lite matches the baseline with a quarter of the weights and beats it on the hardest region (ET, 0.809 vs 0.793), and UNet-UltraLite holds 0.854 with 3% of the baseline's parameters. Accuracy only breaks down at UNet-Tiny, two orders of magnitude below baseline. The conventional configuration is not merely inefficient; on this dataset it is not even the most accurate.

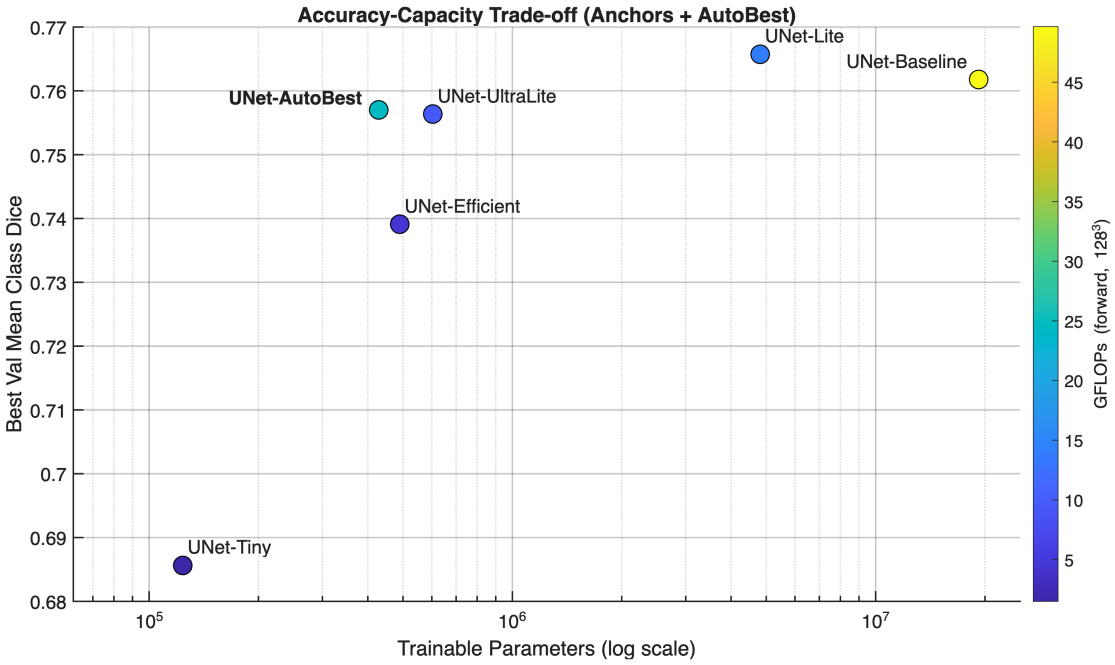


Figure 1. Accuracy against capacity, colour-coded by forward-pass GFLOPs. The searched model occupies a region of the trade-off that no hand-designed anchor reaches.

4 Cost-aware architecture search

The search maximises a scalar objective that trades accuracy against two independent notions of cost — storage (parameters) and computation (FLOPs), which do not move together for 3D convolutions:

$$\text{score}(a) = \text{Dice}(a) - \lambda \cdot \log(\text{params}(a)/10^6 + 1) - \mu \cdot \log(\text{GFLOPs}(a) + 1)$$

Logarithms are used so that the penalty is scale-free: halving a model matters equally whether it starts at 20M or 200K parameters. The search space covers channel width (multipliers 0.70–1.40 applied to the current spec, rounded to multiples of 8 and kept strictly increasing), encoder depth (3–5 levels) and residual units per block (0–2), with strides fixed at 2 between levels. Candidates are explored by local moves from an incumbent and ranked on short proxy training runs.

4.1 The penalty weights are not free parameters

λ and μ determine the answer entirely, so choosing them by intuition would make the search a restatement of the designer's prior. Instead the whole procedure is run at every point of a 5×5 grid, $\lambda \in \{0.05, 0.10, 0.15, 0.20, 0.30\} \times \mu \in \{0.02, 0.05, 0.10, 0.15, 0.20\}$, and the resulting surface is inspected before committing.

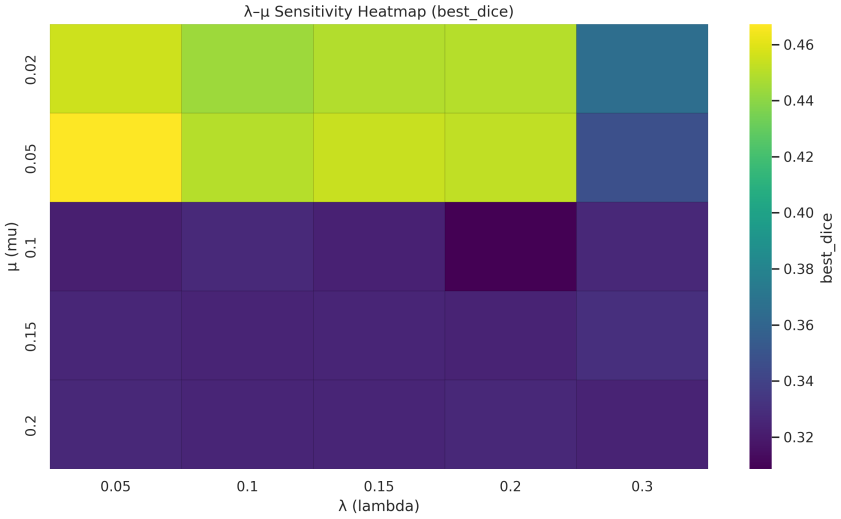


Figure 2. Proxy Dice of the best architecture found at each (λ, μ). The lower three rows are flat: the FLOP penalty has saturated and the objective no longer discriminates.

The surface is more informative than the winning cell. For $\mu \geq 0.10$ every cell collapses to proxy Dice ≈ 0.32 and the search returns the same network, (8,16,32,64,120), regardless of λ. In that regime the FLOP term swamps the accuracy term and the search has effectively stopped optimising for segmentation quality — a failure mode that would be invisible had a single (λ, μ) been fixed in advance. Useful behaviour is confined to $\mu \leq 0.05$, and within that band λ has a mild effect until 0.30, where the parameter penalty begins to bite. The dense search was therefore run at $\lambda = 0.05$, $\mu = 0.02$.

That run evaluated 42 candidates over six iterations of local moves. The top ten were retrained at higher fidelity and re-ranked, since proxy Dice is reliable for shortlisting but not for final ordering. The winner:

```
{"channels": [24, 32, 80], "strides": [2, 2], "num_res_units": 2}
```

This is a three-level encoder. Every hand-designed anchor in the sweep used four or five, so the search did not simply shrink a human design — it changed the depth/width balance in a direction that was not on the manual grid. The search implementation emits a standalone, runnable training script for the winning specification rather than a configuration file, so the selected model is reproducible without re-running the search.

5 Depthwise-separable compression

The searched topology was then compressed by replacing every 3D convolution inside the encoder and decoder residual blocks with a depthwise convolution (one filter per input channel) followed by a pointwise 1×1×1 convolution. Split, pipeline, loss, schedule and augmentation were held fixed, so the resulting change is attributable to the operator factorisation alone.

Parameters fall from 428,553 to 100,477 and cost from 24.6 to 5.5 GFLOPs — 191× smaller and 9× cheaper than the baseline — at 0.813 mean region Dice, a drop of 0.039. The degradation is not uniform: WT loses 0.024 while ET loses 0.042, which is consistent with the enhancing core being the

region that most needs cross-channel mixing at full rank. For a deployment that mainly needs tumour extent, the compressed model is close to free; for one that needs accurate enhancing-tumour delineation, it is not.

6 Uncertainty estimation

A compressed model that is deployed without supervision needs to say where it is unsure. Dropout was inserted into the compressed architecture and kept stochastic at inference; twenty forward passes per case give a predictive distribution, and per-region predictive Bernoulli entropy (normalised by log 2) forms the uncertainty map. Evaluation follows the QU-BraTS protocol [4], reimplemented here:

```
score = AUC(Dice vs T) + (1 − AUC(FTP vs T)) + (1 − AUC(FTN vs T))
```

where T sweeps uncertainty percentiles and progressively removes the least confident voxels. A useful uncertainty map raises Dice as uncertain voxels are filtered out without discarding correct predictions; a perfect score is 3.

Metric	Value	Interpretation
Mean region Dice (MC mean)	0.818	prediction quality of the sampled model
QU-BraTS score (mean)	2.650	out of 3; WT 2.753 / TC 2.630 / ET 2.568
Expected calibration error	0.011	predicted probabilities are well calibrated
Uncertainty–error Spearman, voxel	0.224	entropy tracks error within a scan
Uncertainty–error Spearman, case	0.114	entropy does not identify failed cases

Table 2. MC-dropout uncertainty over the 74 validation cases, T = 20 passes.

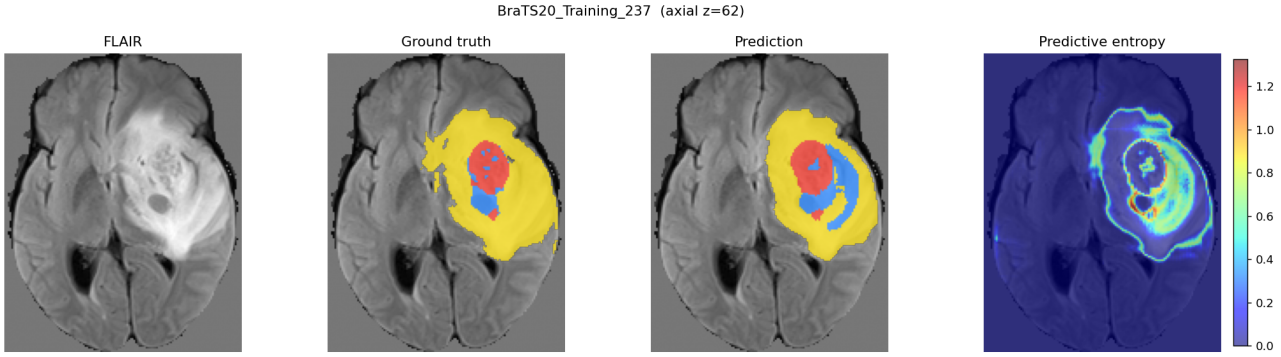


Figure 3. A validation case: FLAIR input, ground truth, prediction, and per-voxel predictive entropy. Uncertainty concentrates on the tumour boundary and the enhancing/necrotic interface — where the errors are.

Calibration is good and the qualitative behaviour is what one wants: entropy is near zero through healthy parenchyma and rises sharply at the boundaries the model gets wrong. The case-level result is the honest caveat. A voxel-level correlation of 0.22 with a case-level correlation of only 0.11 means that aggregate uncertainty over a scan is a poor predictor of whether that scan was segmented badly. Averaging entropy over a volume is dominated by boundary voxels, which are numerous and uncertain in every case, including the well-segmented ones. These maps therefore support directing a radiologist's attention within a study; they do not support automatically flagging a study for review. This is a known limitation of MC dropout rather than an implementation defect, and it is worth stating plainly because the headline QU-BraTS number does not reveal it.

7 Limitations

- Single split, single seed. There is no cross-validation, so the ~0.005 Dice separating the top four models is within run-to-run noise. The table should be read as evidence about orders of magnitude in model size, not as a ranking of the top entries.
- Validation is not a test set. All numbers come from a held-out 20% of the BraTS 2020 training set and are not comparable to official leaderboard scores computed on the hidden test set.
- Proxy fidelity. Search candidates are ranked after a handful of epochs; proxy Dice (0.3–0.55) is far below converged Dice (~0.85). The top-10 rerank mitigates this, but the search could still be

discarding an architecture that is a late bloomer.

- One compression axis. Depthwise separability was applied post hoc rather than being included in the search space. Searching over operator type directly would be the natural next step.
- One uncertainty method. MC dropout is a cheap approximation; deep ensembles typically calibrate better and would be the reference point for a fair claim about uncertainty quality.

8 Conclusion

Treating model size as a search variable rather than a convention produced a 3D U-Net that is 45× smaller than the standard configuration at effectively unchanged accuracy, and a further factorisation reached 191× smaller at a measurable but bounded cost. The methodological point worth carrying forward is smaller than the headline number: sweeping the penalty weights rather than fixing them revealed a large region of the objective's hyperparameter space in which the search silently stops working. Reporting that surface is what makes the selected operating point defensible.

On the uncertainty side, MC dropout on the compressed model is well calibrated and localises error within a scan, but does not identify which scans failed. That distinction determines what such a system could responsibly be used for, and it is not visible in the aggregate score.

Code, configuration and all experiment records are available at `github.com/tahsinaydin02/btseg-autobest-unet` under the MIT licence. The BraTS 2020 dataset is not redistributed.

References

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